



CS344: Operating Systems Lab

Lab Quiz, October 23, 2024

Set - D

Indian Institute of Technology Guwahati

Duration: 40 Minutes

Total : 20 Marks

Name: _____

Roll No.: _____

Student Signature: _____

Invigilator Signature: _____

(10 Questions)

Part - A

[10 Marks]

> Please mention only the option you select inside the box in capital letters.

Q1. What is the significance of the address 0x7c00 during the boot process?

- a> It is where the BIOS is stored.
- b> It is the starting address of the kernel.
- c> It is the location where the boot sector is loaded into memory.
- d> It is the default address for user programs.

Q2. Which of the following best describes the role of wait2() of the assignment?

- a> It tracks the process execution states but does not return any time information to the user program.
- b> It calculates the cumulative time spent by a process in the SLEEPING, RUNNABLE, and RUNNING states.
- c> It only tracks the total runtime of the process.
- d> It waits for all child processes to finish and then returns the total sleep time of the system.

Q3. Which of the following statements is true about context switching

in xv6?

- a>** Context switching only happens when a process finishes execution.
- b>** Context switching saves the state of the current process and loads the state of another process.
- c>** Context switching is performed by user-level threads.
- d>** Context switching does not involve saving the process state.

Q4. Which of the following is a key feature of paging in operating systems?

- a>** Memory is allocated contiguously for all processes.
- b>** Processes are limited to using only available physical memory.
- c>** Memory can be divided into fixed-size pages that are swapped in and out of physical memory.
- d>** Memory can be divided into fixed-size pages that are swapped in and out of physical memory.

Q5. In the modified sbrk() function, what is tricked by incrementing proc->sz without actual allocation?

- a>** The page table.
- b>** The process into thinking it has the requested memory.
- c>** The CPU scheduler.
- d>** The trap handler.

Q6. What does the sys_draw system call do when the provided buffer is too small to hold the ASCII art image?

- a>** It retries with a larger buffer.
- b>** It returns a positive value.
- c>** It crashes the system.
- d>** It returns a negative value.

Q7. Which system call returns information about processes, including

their pid, ticks, and tickets?

- a> getinfo().
- b> processinfo().
- c> getprocessesinfo().
- d> ps().

Q8. What does the p->pgdir field in the struct proc represent in xv6?

- a> The process's execution state (e.g., running, waiting, exiting).
- b> The pointer to the process's user stack.
- c> The process's page table, used for address translation.
- d> The priority level of the process.

Q9. Which fields are to be added to the proc struct for performance evaluation?

- a> ppid, pid, name.
- b> ctime, stime, retime, runtime.
- c> start_time, end_time, total_time.
- d> process_id, user_id, status.

Q10. What is the function of the proc.h file in xv6?

- a> It defines the structure for system calls.
- b> It contains the code for the lottery scheduler.
- c> It defines the process control block (struct proc).
- d> It defines utility functions like argptr().

Q1. Which of the following accurately describes the role of the ZOMBIE state in xv6?

- a> A process in the ZOMBIE state is still actively running but waiting for resources.
- b> A process in the ZOMBIE state has terminated but its parent has not yet called wait() to collect its exit status.
- c> A process in the ZOMBIE state is suspended and can be resumed at a later time.
- d> A process in the ZOMBIE state is paused due to an error and can only be restarted after debugging.

Q2. What instruction does the BIOS execute first after the PC is powered on ?

- a> mov instruction to initialize registers.
- b> jmp instruction to jump to a different location in BIOS.
- c> call instruction to invoke a subroutine.
- d> nop instruction to skip to the next step.

Q3. In the context of lazy memory allocation implemented in xv6, what is the main reason for receiving a trap 14 (page fault) when \ running simple commands like echo hi?

- a> The sbrk() system call allocates more memory than required.
- b> Physical memory gets exhausted when the command is run.
- c> The memory requested by sbrk() is not allocated until it is accessed.
- d> The process exceeds its virtual memory limit.

Q4. In the xv6 swapping mechanism, how can the kernel handle the potential race condition that occurs when a page fault handler tries to swap in a page that is already being swapped out by another kernel process?

- a> By using a blocking semaphore to suspend the page fault handler until the swapping-out process finishes.
- b> By employing a busy-wait loop in the page fault handler to continuously check the status of the page until it is swapped back in.
- c> By introducing a reference count for each page, allowing the kernel

- to wait for the page to become available if it's currently being swapped out.
- d>** By terminating the page fault handler and rescheduling the process when the page is already in the process of being swapped out.

Q5. When adding the settickets() system call, which of the following steps ensures that children processes inherit the correct number of tickets after a fork() operation?

- a>** Modifying the fork() system call to copy the parent's ticket count to the child process.
- b>** Automatically setting child processes to default tickets (10) without copying the parent's tickets.
- c>** Making changes in sysproc.c to call settickets() after every fork().
- d>** The child process independently requests tickets through settickets() during creation.

Q6. In order to safely access and modify the process table (ptable), which of the following should be done before reading or writing to it in a system call like getprocessesinfo()?

- a>** Acquire ptable.lock before accessing the table.
- b>** Interrupt the running process.
- c>** Set process priority to avoid interruptions.
- d>** Disable interrupts.

Q7. Which kernel organization allows most of the operating system to run in user mode while minimizing the code running in kernel mode?

- a>** Monolithic kernel.
- b>** Microkernel.
- c>** Hybrid kernel.
- d>** Exokernel.

Q8. When implementing the user program to display animation in, what

file should you modify to ensure the binary is automatically included in the filesystem of xv6 upon running make qemu??

- a>** user/sleep.c.
- b>** Makefile.
- c>** user/echo.c.
- d>** User/ulib.c.

Q9. What is the default scheduling policy used in the xv6 operating system ?

- a>** Priority-based scheduling.
- b>** First-Come, First-Served (FCFS).
- c>** Round-robin scheduling.
- d>** Lottery scheduling.

Q10. Which file in xv6 contains the array of function pointers for system calls?

- a>** trap.c.
- b>** bootmain.c.
- c>** sysproc.c.
- d>** syscall.c.